

Exam 1 SG

Sunday, September 20, 2026 3:31 PM

1. Nomenclature, Taxonomy, & Classification

Plant Systematics vs. Taxonomy:

- - **Taxonomy:** Description, identification, nomenclature (formal naming of taxa), and classification (grouping of entities).
 - **Plant Systematics:** Encompasses taxonomy *plus* reconstructing evolutionary history using phylogenetics.

Characters & Character

States: •

Character: An observable biological attribute or variable (e.g., petal shape, stamen

number, nucleotide position in a DNA sequence).

Character State: The specific condition or expression of that character in an organism

(e.g., 0 vs. 1, linear vs. ovate, A vs. G).

Scientific Names & Binomial Nomenclature:

- - Follows the format: *Genus* + *specific epithet*. Both words are italicized (or underlined if handwritten). The genus is always capitalized (noun-like); the specific epithet is lowercase (adjective-like).
 - **Authority:** The person who first described and validly published the species (e.g., *Quercus alba* L., where "L." denotes Linnaeus). The authority is plain text (never italicized). If a taxon is transferred to a new genus, the original describing author is enclosed in parentheses, followed by the transferring author.

ICN (International Code of Nomenclature) Rules for Naming Species:

- 1. Designation of a physical **type** specimen.
 2. Publication in primary, peer-reviewed scientific literature.
 3. Formal description in English or Latin.

Ranks vs. Position:

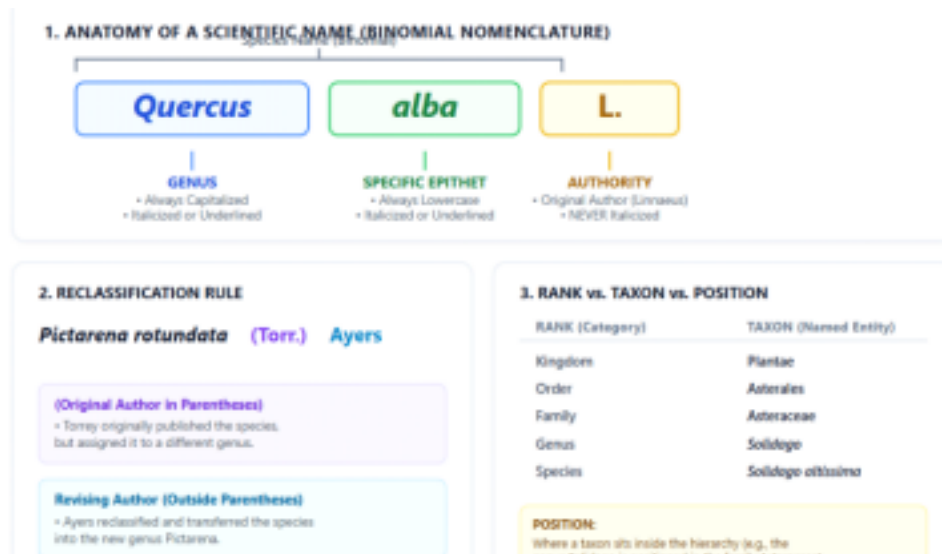
- - **Rank:** The categorical hierarchical level (Domain, Kingdom, Phylum/Division, Class, Order, Family, Genus, Species). Taxa in the same family must share the same order, but need not share the same genus.
 - **Position:** The specific placement of a taxon within higher-ranking taxonomic groups.

• Key points :

- Species names include:
 - Genus
 - Specific epithet

- Authority
- Nomenclature follows a strict set of rules
 - Species must be formally named and described
 - Classification reflects evolutionary history
 - Derived from phylogenies
 - Subject to change

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2. Land Plants & Key Plant Families

Land Plants (Embryophytes):

- Evolved from a shared green algal common ancestor.
Defined by adaptations for life on land, including a waxy **cuticle** to prevent water loss and protected multicellular embryos. Red algae are photosynthetic eukaryotes within Archaeplastida, but lack land plant embryophytic adaptations.
Major groups: Bryophytes, Lycophytes, Monilophytes (ferns & allies), Gymnosperms, and Angiosperms (flowering plants).

Key Points

- Groups of plants are defined by shared characteristics
- Systematics has two goals:
 - Taxonomy: the description, identification, classification, and naming of plants
 - Linnaeus designed our system
 - Describing the evolutionary history of plants through phylogenies

Asteraceae (Sunflower / Composite Family)- many flowers not just one

Head / Capitulum: A specialized inflorescence (group of flowers) that mimics a single flower

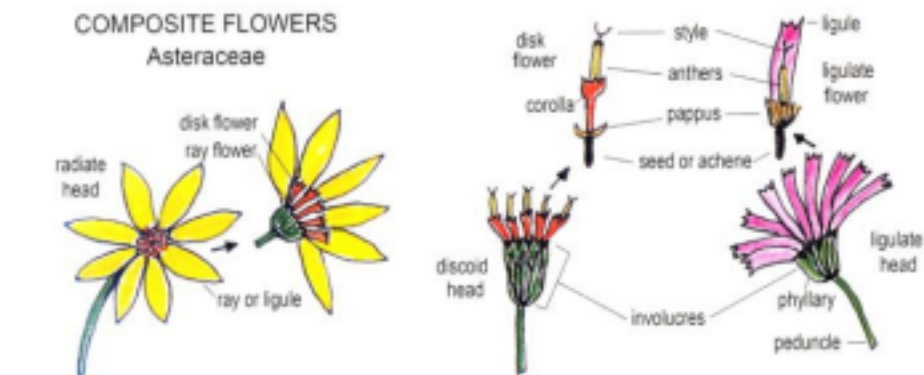
- but is composed of many tiny **florets (tiny flowers)** attached to a common **receptacle**.
- **Involucre & Phyllaries:** The base of the head/capitulum
- **Pappus:** A modified calyx (sepals) reduced to bristles, hairs, scales, or awns sitting atop the ovary/achene fruit to aid dispersal. Simple terms: The calyx is modified to be fluffy or spiky.

Floret Types:

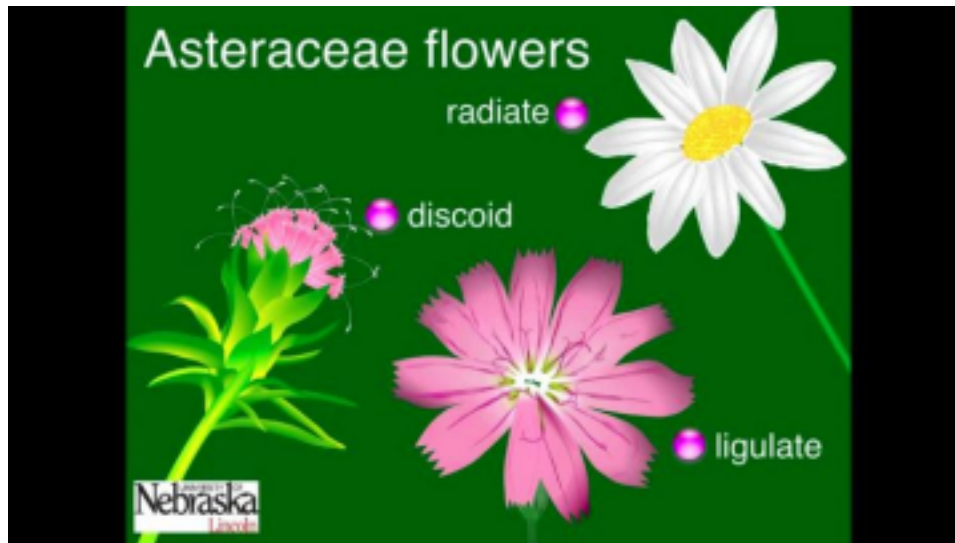
- **Ray floret:** One strappy petal w/ **3 lobes/teeth** at the tip.
- **Disk floret:** tubular corolla of fused petals ending in **5 small lobes**.
- **Ligulate floret:** strap-shaped floret with **5 lobes/teeth** at the tip.

Head Types:

- **Radiate head:** Features central disk florets surrounded by an outer ring of ray florets (e.g., daisies, sunflowers, *Solidago*).
- **Discoid head:** Contains exclusively disk florets.
- **Ligulate head:** Contains exclusively ligulate florets (e.g., dandelions).



Thanks to James Lightner for allowing the use of this image; you can find this and other botanical terms illustrated in the book, *San Diego County Native Plants* by James Lightner (2011).



Key points

:

- Asteraceae have a characteristic head of florets
- The calyx is modified to be fluffy or spiky
- Disk florets, ray florets, and ligulate florets
- Asteraceae is a big family with many fall-blooming species
- Poaceae, Cyperaceae, Juncaceae are graminoids
- Tiny flowers
- Long, thin leaves
- Graminoids are important species in some habitats

The Graminoids (Grass-like Monocots)

Note: "Graminoid" is an informal morphological descriptor, not a formal taxonomic family name (hence uncapitalized).

Feature Poaceae (Grass Family) Cyperaceae (Sedge Family) Juncaceae (Rush Family) Plant Sy_ Page 3

Feature Poaceae (Grass Family) Cyperaceae (Sedge Family)

Stem Round, hollow,
jointed/noded ("reeds have

joints")
3-sided (triangular), solid
("sedges have edges")

Leaves & Sheaths

stem; distinct **ligule** at
junction of blade and
sheath

Inflorescence & Perianth

Open sheath wrapping stigmas and dangling

Tiny florets in
spikelets; feathery

Juncaceae (Rush Family)

Round, solid in the middle
("rushes are round")

3-ranked leaves;
closed sheath

Tiny flowers with
regular perianth of **6**
chaffy tepals; 3

Tiny florets in
spikelets; **no perianth**
Basal or 3-ranked

Fruit Grain (caryopsis) Achene Round **capsule**

3. Tree Thinking & Phylogenetics

Phylogenetic Tree Anatomy:

- **Lineage:** An ancestor-descendant sequence of populations through time.
- **Node:** A branching point indicating a lineage splitting/speciation event; represents the most recent **common ancestor** of the descendant clades.
- **Internode:** Branch segment between two successive nodes.
- **Root / Base:** Represents the deepest ancestral lineage; tips represent modern terminal taxa.

Clades & Group Typology:

- **Monophyletic Group (Clade):** A common ancestor and **all** of its descendants.
- **Paraphyletic Group:** A common ancestor and **some, but not all**, descendants (e.g., traditional *Errazurizia* or *Coelogyne*).
- **Polyphyletic Group:** A grouping that unites descendants from more than one separate common ancestor.

Character Trait

Evolution: •

Ancestral / Plesiomorphic (Symplesiomorphy): Character state inherited from the tree's ancestral lineage.

Derived / Apomorphic: A newly evolved character state.

Synapomorphy: A shared, derived homologous character state uniting two or more

- taxa.
- **Autapomorphy:** A derived character state unique to a single terminal branch/taxon.

Homology vs. Homoplasy:

Homology: Morphological or genetic similarity resulting directly from shared ancestry, evolved in common ancestor, inherited by lineages that came from it. **Homoplasy:** Similarity not due to common descent, generated by:

Convergence: Independent origin of similar states in distinct lineages under similar

- selective pressures, where distantly related or unrelated plant lineages independently evolve similar physical, physiological, or metabolic traits in response to comparable environmental pressure
- **Reversal:** An evolutionary transition back to an ancestral character state.

Outgroup: A taxon chosen outside the ingroup to root the tree and polarize characters

(determining which state is ancestral vs. derived).

Reticulation & Hybridization:

- Standard cladograms assume purely bifurcating (splitting) evolution.
- **Hybridization** merges lineages, causing reticulation (network-like reconnecting branches).

Mapping Characters onto trees

- Outgroup has ancestral character states, only map transitions to derived states Plant Sy_ Page 4

- Outgroup has ancestral character states, only map transitions to derived states
- Be parsimonious- map smallest # of character state changes

- - Map convergence + reversal
 - Sometimes can't tell what it is

Branch length can indicate # of evolutionary changes proportional to the # of evolutionary changes that led to branch

- Usually base pair substitutions
- Scale bar: # or # per base pair

• Key points:

- When two taxa are similar due to common ancestry it's homology •
- A synapomorphy is a novel, derived character state that occurs in two or more taxa
- When two taxa are similar due to reversal or convergent evolution it's homoplasy
- We can map character state transitions onto a phylogeny to understand their evolution
- We use parsimony to map character states on a tree
 - The explanation with the fewest transitions is the best
 - The outgroup establishes which traits are ancestral

4. Molecular Data & Tree Construction

Molecular

Characters: •

Nucleotide Base: Individual positions in an aligned sequence act as characters (character

- states = A, C, T, G). Sites where all taxa share the same base are phylogenetically uninformative.

- **Indel:** An insertion or deletion event in sequence alignment.

Selecting Genomes & DNA Regions:

Chloroplast DNA (cpDNA): E.g., *rbcL*, *matK*, *ndhF*. Maternally inherited in most

- angiosperms, non-recombining, slower evolutionary rates; ideal for deeper (family/order level) phylogenetic resolution.

Nuclear DNA (nrDNA): E.g., ITS cistron. Biparentally inherited, faster evolving; suited for

- shallow, recent species-level relationships and uncovering hybrid/reticulate events.

Angiosperms353: Universal target-capture panel sequencing 353 single-copy nuclear

- protein-coding genes to resolve deep and shallow angiosperm clades.

Methods of Phylogeny

Reconstruction: •

Parsimony Analysis: Employs Occam's razor; identifies the tree that requires the minimal

1. total number of evolutionary transitions (steps).

- 2. **Maximum Likelihood (ML):** Uses specific probabilistic models of base substitution rates to identify the tree topology that maximizes the probability of observing the sequence data.
- 3. **Bayesian Analysis:** Evaluates posterior probability distributions using MCMC algorithms to find the most probable tree given the data and model.
- 4. **ASTRAL Analysis:** Coalescent species-tree method that reconciles conflicting individual gene trees, explicitly handling incomplete lineage sorting (ILS).

5. Branch Lengths, Robustness, & Consensus Trees

Branch Length Interpretation:

- **Cladogram:** Branch lengths are arbitrary; conveys only relative branching topology.
- **Phylogram:** Branch lengths represent evolutionary divergence (either absolute count of nucleotide changes or substitutions per site calibrated to a scale bar).

Cladogram Robustness

Metrics: •

- **Bootstrapping (Parsimony & ML):** Resampling columns of alignment data with replacement to generate hundreds of pseudo-replicate datasets. The bootstrap value indicates the percentage of replicated trees recovering that specific clade. Values represent strong statistical support.

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represent strong statistical support.

- **Posterior Probabilities (Bayesian):** Statistical probabilities between 0.0 and 1.0 that a clade is true given the model and data. Values (**or**) indicate strong support.
- **Proportion of Supporting Genes (ASTRAL):** The proportion of individual gene trees that agree with the species clade. Values (**50%**) are considered acceptable/good support; indicates weak support.

Polytomies & Consensus

Trees: •

- **Polytomy:** A node with more than two immediate descendant branches, signaling uncertainty/insufficient data or rapid simultaneous divergence.
- **Strict Consensus Tree:** When multiple equally parsimonious trees exist, this summary tree collapses conflicting nodes into polytomies, preserving only the clades present in 100% of the candidate trees.

• Key points :

- A phylogeny is a hypothesis that shows how taxa are related
- Shared branches and nodes represent a shared common ancestor
- Phylogenies tell us about the evolution of characters
- A set of species derived from one common ancestor is a monophyletic group

- Named taxa should be monophyletic groups
- Hybridization creates a reticulate phylogeny
- Key points:
 - Multiple methods for constructing phylogenies
 - Maximum Parsimony
 - Maximum Likelihood (ML)
 - ASTRAL analysis
 - Bayesian analysis
 - Measures of clade robustness (accuracy)
 - Bootstrapping with Parsimony and ML
 - Greater than 70% is good
 - Posterior probabilities with Bayesian
 - Greater than 90% or 0.9 is good
 - Clades that aren't robust may be collapsed into a polytomy on a consensus tree